

4

Performance Specifications

The most important objective of a control system is to achieve certain performance specifications in addition to providing the internal stability. One way to describe the performance specifications of a control system is in terms of the size of certain signals of interest. For example, the performance of a tracking system could be measured by the size of the tracking error signal. In this chapter, we look at several ways of defining a signal's size, i.e., at several norms for signals. Of course, which norm is most appropriate depends on the situation at hand. For that purpose, we shall first introduce some normed spaces and some basic notions of linear operator theory, in particular, the Hardy spaces $\mathcal{H}_2$ and $\mathcal{H}_\infty$ are introduced. We then consider the performance of a system under various input signals and derive the worst possible outputs with the class of input signals under consideration. We show that $\mathcal{H}_2$ and $\mathcal{H}_\infty$ norms come out naturally as measures of the worst possible performance for many classes of input signals. Some state space methods of computing real rational $\mathcal{H}_2$ and $\mathcal{H}_\infty$ transfer matrix norms are also presented.

4.1 Normed Spaces

Let V be a vector space over $\mathbb{C}$ (or $\mathbb{R}$) and let $\|\cdot\|$ be a norm defined on V . Then V is a *normed space*. For example, the vector space $\mathbb{C}^n$ with any vector p -norm, $\|\cdot\|_p$, for $1 \leq p \leq \infty$, is a normed space. As another example, consider the linear vector space $C[a, b]$ of all bounded continuous functions on the real interval $[a, b]$. Then $C[a, b]$

becomes a normed space if a *supremum norm* is defined on the space:

$$\|f\|_\infty := \sup_{t \in [a, b]} |f(t)|.$$

A sequence $\{x_n\}$ in a normed space V is called a *Cauchy sequence* if $\|x_n - x_m\| \rightarrow 0$ as $n, m \rightarrow \infty$. A sequence $\{x_n\}$ is said to *converge* to $x \in V$, written $x_n \rightarrow x$, if $\|x_n - x\| \rightarrow 0$. A normed space V is said to be *complete* if every Cauchy sequence in V converges in V . A complete normed space is called a *Banach space*. For example, $\mathbb{R}^n$ and $\mathbb{C}^n$ with the usual spatial p -norm, $\|\cdot\|_p$ for $1 \leq p \leq \infty$, are Banach spaces. (One should not be confused with the notation $\|\cdot\|_p$ used here and the same notation used below for function spaces because usually the context will make the meaning clear.)

The following are some more examples of Banach spaces:

$l_p[0, \infty)$ spaces for $1 \leq p < \infty$:

For each $1 \leq p < \infty$, $l_p[0, \infty)$ consists of all sequences $x = (x_0, x_1, \dots)$ such that $\sum_{i=0}^{\infty} |x_i|^p < \infty$. The associated norm is defined as

$$\|x\|_p := \left(\sum_{i=0}^{\infty} |x_i|^p \right)^{1/p}.$$

$l_\infty[0, \infty)$ space:

$l_\infty[0, \infty)$ consists of all bounded sequences $x = (x_0, x_1, \dots)$, and the l_∞ norm is defined as

$$\|x\|_\infty := \sup_i |x_i|.$$

$\mathcal{L}_p(I)$ spaces for $1 \leq p \leq \infty$:

For each $1 \leq p < \infty$, $\mathcal{L}_p(I)$ consists of all Lebesgue measurable functions $x(t)$ defined on an interval $I \subset \mathbb{R}$ such that

$$\|x\|_p := \left(\int_I |x(t)|^p dt \right)^{1/p} < \infty, \text{ for } 1 \leq p < \infty$$

and

$$\|x(t)\|_\infty := \text{ess sup}_{t \in I} |x(t)|.$$

Some of these spaces, for example, $\mathcal{L}_2(-\infty, 0]$, $\mathcal{L}_2[0, \infty)$ and $\mathcal{L}_2(-\infty, \infty)$, will be discussed in more detail later on.

$C[a, b]$ space:

$C[a, b]$ consists of all continuous functions on the real interval $[a, b]$ with the norm defined as

$$\|x\|_\infty := \sup_{t \in [a, b]} |x(t)|.$$

Note that if each component or function is itself a vector or matrix, then the corresponding Banach space can also be formed by replacing the absolute value $|\cdot|$ of each component or function with its spatially normed component or function. For example, consider a vector space with all sequences in the form of

$$x = (x_0, x_1, \dots)$$

where each component x_i is a $k \times m$ matrix and each element of x_i is bounded. Then x_i is bounded in any matrix norm, and the vector space becomes a Banach space if the following norm is defined

$$\|x\|_\infty := \sup_i \phi_i(x_i)$$

where $\phi_i(x_i) := \|x_i\|$ is any matrix norm. This space will also be denoted by l_∞ .

Let V_1 and V_2 be two vector spaces and let T be an operator from $S \subset V_1$ into V_2 . An operator T is said to be *linear* if for any $x_1, x_2 \in S$ and scalars $\alpha, \beta \in \mathbb{C}$, the following holds:

$$T(\alpha x_1 + \beta x_2) = \alpha(Tx_1) + \beta(Tx_2).$$

Moreover, let V_0 be a linear subspace in V_1 . Then the operator $T_0 : V_0 \mapsto V_2$ defined by $T_0 x = Tx$ for every $x \in V_0$ is called the *restriction* of T to V_0 and is denoted as $T|_{V_0} = T_0$. On the other hand, a linear operator $T : V_1 \mapsto V_2$ coinciding with T_0 on $V_0 \subset V_1$ is called an *extension* of T_0 . For example, let $V_0 := \left\{ \begin{bmatrix} x_1 \\ 0 \end{bmatrix} : x_1 \in \mathbb{C}^n \right\} \subset V_1 = \mathbb{C}^{n+m}$, and let

$$T = \begin{bmatrix} A_1 & A_2 \\ 0 & A_3 \end{bmatrix} \in \mathbb{C}^{(n+m) \times (n+m)}, \quad T_0 = \begin{bmatrix} A_1 & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{C}^{(n+m) \times (n+m)}.$$

Then $T|_{V_0} = T_0$.

Definition 4.1 Two normed spaces V_1 and V_2 are said to be *linearly isometric*, denoted by $V_1 \cong V_2$, if there exists a linear operator T of V_1 onto V_2 such that

$$\|Tx\| = \|x\|$$

for all x in V_1 . In this case, the mapping T is said to be an *isometric isomorphism*.

4.2 Hilbert Spaces

Recall the inner product of vectors defined on a Euclidean space $\mathbb{C}^n$:

$$\langle x, y \rangle := x^* y = \sum_{i=1}^n \bar{x}_i y_i \quad \forall x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{C}^n.$$

Note that many important metric notions and geometrical properties such as length, distance, angle, and the energy of physical systems can be deduced from this inner product. For instance, the length of a vector $x \in \mathbb{C}^n$ is defined as

$$\|x\| := \sqrt{\langle x, x \rangle},$$

and the angle between two vectors $x, y \in \mathbb{C}^n$ can be computed from

$$\cos \angle(x, y) = \frac{\langle x, y \rangle}{\|x\| \|y\|}, \quad \angle(x, y) \in [0, \pi].$$

The two vectors are said to be *orthogonal* if $\angle(x, y) = \frac{\pi}{2}$.

We now consider a natural generalization of the inner product on $\mathbb{C}^n$ to more general (possibly infinite dimensional) vector spaces.

Definition 4.2 Let V be a vector space over $\mathbb{C}$. An *inner product* on V is a complex valued function,

$$\langle \cdot, \cdot \rangle : V \times V \longrightarrow \mathbb{C}$$

such that for any $x, y, z \in V$ and $\alpha, \beta \in \mathbb{C}$

$$(i) \quad \langle x, \alpha y + \beta z \rangle = \alpha \langle x, y \rangle + \beta \langle x, z \rangle$$

$$(ii) \quad \langle x, y \rangle = \overline{\langle y, x \rangle}$$

$$(iii) \quad \langle x, x \rangle > 0 \text{ if } x \neq 0.$$

A vector space V with an inner product is called an *inner product space*.

It is clear that the inner product defined above induces a norm $\|x\| := \sqrt{\langle x, x \rangle}$, so that the norm conditions in Chapter 2 are satisfied. In particular, the distance between vectors x and y is $d(x, y) = \|x - y\|$.

Two vectors x and y in an inner product space V are said to be *orthogonal* if $\langle x, y \rangle = 0$, denoted $x \perp y$. More generally, a vector x is said to be orthogonal to a set $S \subset V$, denoted by $x \perp S$, if $x \perp y$ for all $y \in S$.

The inner product and the inner-product induced norm have the following familiar properties.

Theorem 4.1 Let V be an inner product space and let $x, y \in V$. Then

(i) $|\langle x, y \rangle| \leq \|x\| \|y\|$ (*Cauchy-Schwarz inequality*). Moreover, the equality holds if and only if $x = \alpha y$ for some constant α or $y = 0$.

(ii) $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$ (*Parallelogram law*).

(iii) $\|x + y\|^2 = \|x\|^2 + \|y\|^2$ if $x \perp y$.

¹This is the other way round to the usual mathematical convention since we want to have $\langle x, y \rangle = x^* y$ rather than $y^* x$ for $x, y \in \mathbb{C}^n$.

A *Hilbert space* is a complete inner product space with the norm induced by its inner product. Obviously, a Hilbert space is also a Banach space. For example, $\mathbb{C}^n$ with the usual inner product is a (finite dimensional) Hilbert space. More generally, it is straightforward to verify that $\mathbb{C}^{n \times m}$ with the inner product defined as

$$\langle A, B \rangle := \text{Trace } A^* B = \sum_{i=1}^n \sum_{j=1}^m \bar{a}_{ij} b_{ij} \quad \forall A, B \in \mathbb{C}^{n \times m}$$

is also a (finite dimensional) Hilbert space.

Here are some examples of infinite dimensional Hilbert spaces:

$l_2(-\infty, \infty)$:

$l_2(-\infty, \infty)$ consists of the set of all real or complex square summable sequences

$$x = (\dots, x_{-2}, x_{-1}, x_0, x_1, x_2, \dots)$$

i.e.,

$$\sum_{i=-\infty}^{\infty} |x_i|^2 < \infty$$

with the inner product defined as

$$\langle x, y \rangle := \sum_{i=-\infty}^{\infty} \bar{x}_i y_i$$

for $x, y \in l_2(-\infty, \infty)$. The subspaces $l_2(-\infty, 0)$ and $l_2[0, \infty)$ of $l_2(-\infty, \infty)$ are defined similarly and consist of sequences of the form $x = (\dots, x_{-2}, x_{-1})$ and $x = (x_0, x_1, x_2, \dots)$, respectively.

Note that we can also define a corresponding Hilbert space even if each component x_i is a vector or a matrix; in fact, the following inner product will suffice:

$$\langle x, y \rangle := \sum_{i=-\infty}^{\infty} \text{Trace}(x_i^* y_i).$$

$\mathcal{L}_2(I)$ for $I \subset \mathbb{R}$:

$\mathcal{L}_2(I)$ consists of all square integrable and Lebesgue measurable functions defined on an interval $I \subset \mathbb{R}$ with the inner product defined as

$$\langle f, g \rangle := \int_I f(t)^* g(t) dt$$

for $f, g \in \mathcal{L}_2(I)$. Similarly, if the function is vector or matrix valued, the inner product is defined correspondingly as

$$\langle f, g \rangle := \int_I \text{Trace}[f(t)^* g(t)] dt.$$

Some very often used spaces in this book are $\mathcal{L}_2[0, \infty)$, $\mathcal{L}_2(-\infty, 0]$, $\mathcal{L}_2(-\infty, \infty)$. More precisely, they are defined as

$\mathcal{L}_2 = \mathcal{L}_2(-\infty, \infty)$: Hilbert space of matrix-valued functions on $\mathbb{R}$, with inner product

$$\langle f, g \rangle := \int_{-\infty}^{\infty} \text{Trace}[f(t)^* g(t)] dt.$$

$\mathcal{L}_{2+} = \mathcal{L}_2[0, \infty)$: subspace of $\mathcal{L}_2(-\infty, \infty)$ with functions zero for $t < 0$.

$\mathcal{L}_{2-} = \mathcal{L}_2(-\infty, 0]$: subspace of $\mathcal{L}_2(-\infty, \infty)$ with functions zero for $t > 0$.

Let $\mathcal{H}$ be a Hilbert space and $M \subset \mathcal{H}$ a subset. Then the *orthogonal complement* of M , denoted by $\mathcal{H} \ominus M$ or $M^\perp$, is defined as

$$M^\perp = \{x : \langle x, y \rangle = 0, \forall y \in M, x \in \mathcal{H}\}.$$

It can be shown that $M^\perp$ is closed (hence $M^\perp$ is also a Hilbert space). For example, let $M = \mathcal{L}_{2+} \subset \mathcal{L}_2$, then $M^\perp = \mathcal{L}_{2-}$ is a Hilbert space.

Let M and N be subspaces of a vector space V . V is said to be the *direct sum* of M and N , written $V = M \oplus N$, if $M \cap N = \{0\}$, and every element $v \in V$ can be expressed as $v = x + y$ with $x \in M$ and $y \in N$. If V is an inner product space and M and N are orthogonal, then V is said to be the *orthogonal direct sum* of M and N . As an example, it is easy to see that $\mathcal{L}_2$ is the orthogonal direct sum of $\mathcal{L}_{2-}$ and $\mathcal{L}_{2+}$. Similarly, $l_2(-\infty, \infty)$ is the orthogonal direct sum of $l_2(-\infty, 0)$ and $l_2[0, \infty)$.

The following is a version of the so-called *orthogonal projection theorem*:

Theorem 4.2 *Let $\mathcal{H}$ be a Hilbert space, and let M be a closed subspace of $\mathcal{H}$. Then for each vector $v \in \mathcal{H}$, there exist unique vectors $x \in M$ and $y \in M^\perp$ such that $v = x + y$, i.e., $\mathcal{H} = M \oplus M^\perp$. Moreover, $x \in M$ is the unique vector such that $d(v, M) = \|v - x\|$.*

Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two Hilbert spaces, and let A be a bounded linear operator from $\mathcal{H}_1$ into $\mathcal{H}_2$. Then there exists a unique linear operator $A^* : \mathcal{H}_2 \longrightarrow \mathcal{H}_1$ such that for all $x \in \mathcal{H}_1$ and $y \in \mathcal{H}_2$

$$\langle Ax, y \rangle = \langle x, A^*y \rangle.$$

A^* is called the *adjoint* of A . Furthermore, A is called *self-adjoint* if $A = A^*$.

Let $\mathcal{H}$ be a Hilbert space and $M \subset \mathcal{H}$ be a closed subspace. A bounded operator P mapping from $\mathcal{H}$ into itself is called the *orthogonal projection* onto M if

$$P(x + y) = x, \forall x \in M \text{ and } y \in M^\perp.$$

4.3 Hardy Spaces $\mathcal{H}_2$ and $\mathcal{H}_\infty$

Let $S \subset \mathbb{C}$ be an open set, and let $f(s)$ be a complex valued function defined on S :

$$f(s) : S \longrightarrow \mathbb{C}.$$

Then $f(s)$ is said to be *analytic at a point* z_0 in S if it is differentiable at z_0 and also at each point in some neighborhood of z_0 . It is a fact that if $f(s)$ is analytic at z_0 then f has continuous derivatives of all orders at z_0 . Hence, a function analytic at z_0 has a power series representation at z_0 . The converse is also true, i.e., if a function has a power series at z_0 , then it is analytic at z_0 . A function $f(s)$ is said to be *analytic in S* if it has a derivative or is analytic at each point of S . A matrix valued function is analytic in S if every element of the matrix is analytic in S . For example, all real rational stable transfer matrices are analytic in the right-half plane and e^{-s} is analytic everywhere.

A well known property of the analytic functions is the so-called *Maximum Modulus Theorem*.

Theorem 4.3 *If $f(s)$ is defined and continuous on a closed-bounded set S and analytic on the interior of S , then the maximum of $|f(s)|$ on S is attained on the boundary of S , i.e.,*

$$\max_{s \in S} |f(s)| = \max_{s \in \partial S} |f(s)|$$

where ∂S denotes the boundary of S .

Next we consider some frequently used complex (matrix) function spaces.

$\mathcal{L}_2(j\mathbb{R})$ Space

$\mathcal{L}_2(j\mathbb{R})$ or simply $\mathcal{L}_2$ is a Hilbert space of matrix-valued (or scalar-valued) functions on $j\mathbb{R}$ and consists of all complex matrix functions F such that the integral below is bounded, i.e.,

$$\int_{-\infty}^{\infty} \text{Trace}[F^*(j\omega)F(j\omega)] d\omega < \infty.$$

The inner product for this Hilbert space is defined as

$$\langle F, G \rangle := \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{Trace}[F^*(j\omega)G(j\omega)] d\omega$$

for $F, G \in \mathcal{L}_2$, and the inner product induced norm is given by

$$\|F\|_2 := \sqrt{\langle F, F \rangle}.$$

For example, all real rational strictly proper transfer matrices with no poles on the imaginary axis form a subspace (not closed) of $\mathcal{L}_2(j\mathbb{R})$ which is denoted by $\mathcal{RL}_2(j\mathbb{R})$ or simply $\mathcal{RL}_2$.

$\mathcal{H}_2$ Space²

$\mathcal{H}_2$ is a (closed) subspace of $\mathcal{L}_2(j\mathbb{R})$ with matrix functions $F(s)$ analytic in $\text{Re}(s) > 0$ (open right-half plane). The corresponding norm is defined as

$$\|F\|_2^2 := \sup_{\sigma > 0} \left\{ \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{Trace}[F^*(\sigma + j\omega)F(\sigma + j\omega)] d\omega \right\}.$$

It can be shown³ that

$$\|F\|_2^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{Trace}[F^*(j\omega)F(j\omega)] d\omega.$$

Hence, we can compute the norm for $\mathcal{H}_2$ just as we do for $\mathcal{L}_2$. The real rational subspace of $\mathcal{H}_2$, which consists of all strictly proper and real rational stable transfer matrices, is denoted by $\mathcal{RH}_2$.

$\mathcal{H}_2^\perp$ Space

$\mathcal{H}_2^\perp$ is the orthogonal complement of $\mathcal{H}_2$ in $\mathcal{L}_2$, i.e., the (closed) subspace of functions in $\mathcal{L}_2$ that are analytic in the open left-half plane. The real rational subspace of $\mathcal{H}_2^\perp$, which consists of all strictly proper rational transfer matrices with all poles in the open right half plane, will be denoted by $\mathcal{RH}_2^\perp$. It is easy to see that if G is a strictly proper, stable, and real rational transfer matrix, then $G \in \mathcal{H}_2$ and $G^\sim \in \mathcal{H}_2^\perp$. Most of our study in this book will be focused on the real rational case.

The $\mathcal{L}_2$ spaces defined above in the frequency domain can be related to the $\mathcal{L}_2$ spaces defined in the time domain. Recall the fact that a function in $\mathcal{L}_2$ space in the time domain admits a bilateral Laplace (or Fourier) transform. In fact, it can be shown that this bilateral Laplace (or Fourier) transform yields an isometric isomorphism between the $\mathcal{L}_2$ spaces in the time domain and the $\mathcal{L}_2$ spaces in the frequency domain (this is what is called *Parseval's relations* or *Plancherel Theorem* in complex analysis):

$$\mathcal{L}_2(-\infty, \infty) \cong \mathcal{L}_2(j\mathbb{R})$$

$$\mathcal{L}_2[0, \infty) \cong \mathcal{H}_2$$

$$\mathcal{L}_2(-\infty, 0] \cong \mathcal{H}_2^\perp.$$

As a result, if $g(t) \in \mathcal{L}_2(-\infty, \infty)$ and if its Fourier (or bilateral Laplace) transform is $G(j\omega) \in \mathcal{L}_2(j\mathbb{R})$, then

$$\|G\|_2 = \|g\|_2.$$

²The $\mathcal{H}_2$ space and $\mathcal{H}_\infty$ space defined below together with the $\mathcal{H}_p$ spaces, $p \geq 1$, which will not be introduced in this book, are usually called Hardy spaces named after the mathematician G. H. Hardy (hence the notation of $\mathcal{H}$).

³See Francis [1987].

Hence, whenever there is no confusion, the notation of functions in the time domain and in the frequency domain will be used interchangeably.

Define an orthogonal projection

$$P_+ : \mathcal{L}_2(-\infty, \infty) \mapsto \mathcal{L}_2[0, \infty)$$

such that, for any function $f(t) \in \mathcal{L}_2(-\infty, \infty)$, we have $g(t) = P_+ f(t)$ with

$$g(t) := \begin{cases} f(t), & \text{for } t \geq 0; \\ 0, & \text{for } t < 0. \end{cases}$$

In this book, P_+ will also be used to denote the projection from $\mathcal{L}_2(j\mathbb{R})$ onto $\mathcal{H}_2$. Similarly, define P_- as another orthogonal projection from $\mathcal{L}_2(-\infty, \infty)$ onto $\mathcal{L}_2(-\infty, 0]$ (or $\mathcal{L}_2(j\mathbb{R})$ onto $\mathcal{H}_2^\perp$). Then the relationships between $\mathcal{L}_2$ spaces and $\mathcal{H}_2$ spaces can be shown as in Figure 4.1.

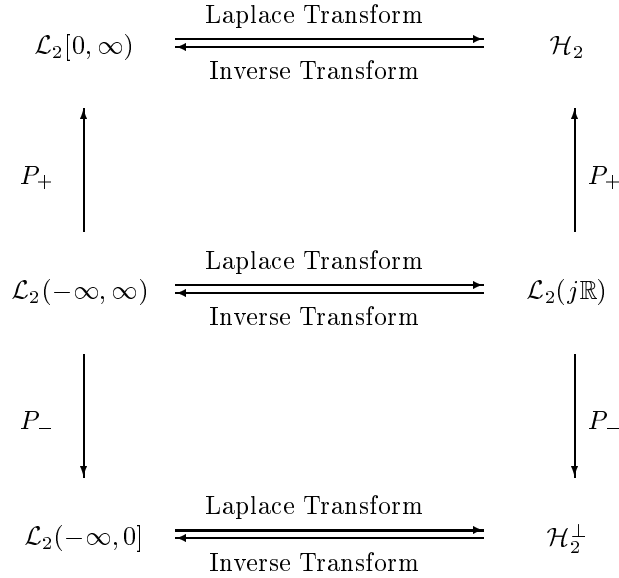


Figure 4.1: Relationships among function spaces

Other classes of important complex matrix functions used in this book are those bounded on the imaginary axis.

$\mathcal{L}_\infty(j\mathbb{R})$ Space

$\mathcal{L}_\infty(j\mathbb{R})$ or simply $\mathcal{L}_\infty$ is a Banach space of matrix-valued (or scalar-valued) functions that are (essentially) bounded on $j\mathbb{R}$, with norm

$$\|F\|_\infty := \operatorname{ess\,sup}_{\omega \in \mathbb{R}} \bar{\sigma}[F(j\omega)].$$

The rational subspace of $\mathcal{L}_\infty$, denoted by $\mathcal{RL}_\infty(j\mathbb{R})$ or simply $\mathcal{RL}_\infty$, consists of all rational proper transfer matrices with no poles on the imaginary axis.

$\mathcal{H}_\infty$ Space

$\mathcal{H}_\infty$ is a (closed) subspace in $\mathcal{L}_\infty$ with functions that are analytic in the open right-half plane and bounded on the imaginary axis. The $\mathcal{H}_\infty$ norm is defined as

$$\|F\|_\infty := \sup_{\operatorname{Re}(s) > 0} \bar{\sigma}[F(s)] = \operatorname{ess\,sup}_{\omega \in \mathbb{R}} \bar{\sigma}[F(j\omega)].$$

The second equality can be regarded as a generalization of the maximum modulus theorem for matrix functions. The real rational subspace of $\mathcal{H}_\infty$ is denoted by $\mathcal{RH}_\infty$ which consists of all proper and real rational stable transfer matrices.

$\mathcal{H}_\infty^-$ Space

$\mathcal{H}_\infty^-$ is a (closed) subspace in $\mathcal{L}_\infty$ with functions that are analytic in the open left-half plane and bounded on the imaginary axis. The $\mathcal{H}_\infty^-$ norm is defined as

$$\|F\|_\infty := \sup_{\operatorname{Re}(s) < 0} \bar{\sigma}[F(s)] = \operatorname{ess\,sup}_{\omega \in \mathbb{R}} \bar{\sigma}[F(j\omega)].$$

The real rational subspace of $\mathcal{H}_\infty^-$ is denoted by $\mathcal{RH}_\infty^-$ which consists of all proper rational transfer matrices with all poles in the open right half plane.

Definition 4.3 A transfer matrix $G(s) \in \mathcal{H}_\infty^-$ is usually said to be *antistable* or *anticausal*.

Some facts about $\mathcal{L}_\infty$ and $\mathcal{H}_\infty$ functions are worth mentioning:

- (i) if $G(s) \in \mathcal{L}_\infty$, then $G(s)\mathcal{L}_2 := \{G(s)f(s) : f(s) \in \mathcal{L}_2\} \subset \mathcal{L}_2$.
- (ii) if $G(s) \in \mathcal{H}_\infty$, then $G(s)\mathcal{H}_2 := \{G(s)f(s) : f(s) \in \mathcal{H}_2\} \subset \mathcal{H}_2$.
- (ii) if $G(s) \in \mathcal{H}_\infty^-$, then $G(s)\mathcal{H}_2^\perp := \{G(s)f(s) : f(s) \in \mathcal{H}_2^\perp\} \subset \mathcal{H}_2^\perp$.

Remark 4.1 The notation for $\mathcal{L}_\infty$ is somewhat unfortunate; it should be clear to the reader that the $\mathcal{L}_\infty$ space in the time domain and in the frequency domain denote completely different spaces. The $\mathcal{L}_\infty$ space in the time domain is usually used to denote signals, while the $\mathcal{L}_\infty$ space in the frequency domain is usually used to denote transfer functions and operators. $\heartsuit$

Let $G(s) \in \mathcal{L}_\infty$ be a $p \times q$ transfer matrix. Then a multiplication operator is defined as

$$\begin{aligned} M_G : \mathcal{L}_2 &\longmapsto \mathcal{L}_2 \\ M_G f &:= Gf. \end{aligned}$$

In writing the above mapping, we have assumed that f has a compatible dimension. A more accurate description of the above operator should be

$$M_G : \mathcal{L}_2^q \mapsto \mathcal{L}_2^p$$

i.e., f is a q -dimensional vector function with each component in $\mathcal{L}_2$. However, we shall suppress all dimensions in this book and assume all objects have compatible dimensions.

It is easy to verify that the adjoint operator $M_G^* = M_{G^*}$.

A useful fact about the multiplication operator is that the norm of a matrix G in $\mathcal{L}_\infty$ equals the norm of the corresponding multiplication operator.

Theorem 4.4 *Let $G \in \mathcal{L}_\infty$ be a $p \times q$ transfer matrix. Then $\|M_G\| = \|G\|_\infty$.*

Remark 4.2 It is also true that this operator norm equals the norm of the operator restricted to $\mathcal{H}_2$ (or $\mathcal{H}_2^\perp$), i.e.,

$$\|M_G\| = \|M_G|_{\mathcal{H}_2}\| := \sup \{\|Gf\|_2 : f \in \mathcal{H}_2, \|f\|_2 \leq 1\}.$$

This will be clear in the proof where an $f \in \mathcal{H}_2$ is constructed. ♥

Proof. By definition, we have

$$\|M_G\| = \sup \{\|Gf\|_2 : f \in \mathcal{L}_2, \|f\|_2 \leq 1\}.$$

First we see that $\|G\|_\infty$ is an upper bound for the operator norm:

$$\begin{aligned} \|Gf\|_2^2 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f^*(j\omega) G^*(j\omega) G(j\omega) f(j\omega) d\omega \\ &\leq \|G\|_\infty^2 \frac{1}{2\pi} \int_{-\infty}^{\infty} \|f(j\omega)\|^2 d\omega \\ &= \|G\|_\infty^2 \|f\|_2^2. \end{aligned}$$

To show that $\|G\|_\infty$ is the least upper bound, first choose a frequency ω_o where $\bar{\sigma}[G(j\omega)]$ is maximum, i.e.,

$$\bar{\sigma}[G(j\omega_o)] = \|G\|_\infty$$

and denote the singular value decomposition of $G(j\omega_o)$ by

$$G(j\omega_o) = \bar{\sigma} u_1(j\omega_o) v_1^*(j\omega_o) + \sum_{i=2}^r \sigma_i u_i(j\omega_o) v_i^*(j\omega_o)$$

where r is the rank of $G(j\omega_o)$ and u_i, v_i have unit length.

If $\omega_o < \infty$, write $v_1(j\omega_o)$ as

$$v_1(j\omega_o) = \begin{bmatrix} \alpha_1 e^{j\theta_1} \\ \alpha_2 e^{j\theta_2} \\ \vdots \\ \alpha_q e^{j\theta_q} \end{bmatrix}$$

where $\alpha_i \in \mathbb{R}$ is such that $\theta_i \in [-\pi, 0)$ and q is the column dimension of G . Now let $\beta_i \geq 0$ be such that

$$\theta_i = \angle \left(\frac{\beta_i - j\omega_0}{\beta_i + j\omega_0} \right),$$

and let f be given by

$$f(s) = \begin{bmatrix} \alpha_1 \frac{\beta_1 - s}{\beta_1 + s} \\ \alpha_2 \frac{\beta_2 - s}{\beta_2 + s} \\ \vdots \\ \alpha_q \frac{\beta_q - s}{\beta_q + s} \end{bmatrix} \hat{f}(s)$$

where a scalar function $\hat{f}$ is chosen so that

$$|\hat{f}(j\omega)| = \begin{cases} c & \text{if } |\omega - \omega_o| < \epsilon \text{ or } |\omega + \omega_o| < \epsilon \\ 0 & \text{otherwise} \end{cases}$$

where ϵ is a small positive number and c is chosen so that $\hat{f}$ has unit 2-norm, i.e., $c = \sqrt{\pi/2\epsilon}$. This in turn implies that f has unit 2-norm. Then

$$\begin{aligned} \|Gf\|_2^2 &\approx \frac{1}{2\pi} \left[\bar{\sigma} [G(-j\omega_o)]^2 \pi + \bar{\sigma} [G(j\omega_o)]^2 \pi \right] \\ &= \bar{\sigma} [G(j\omega_o)]^2 = \|G\|_\infty^2. \end{aligned}$$

If $\omega_0 = \infty$, then $G(j\omega_0)$ is real and v_1 is real. In this case, the conclusion follows by letting $f(s) = v_1 \hat{f}(s)$. $\square$

4.4 Power and Spectral Signals

In this section, we introduce two additional classes of signals that have been widely used in engineering. These classes of signals have some nice statistical and frequency domain representations. Let $u(t)$ be a function of time. Define its autocorrelation matrix as

$$R_{uu}(\tau) := \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T u(t + \tau) u^*(t) dt,$$

if the limit exists and is finite for all τ . It is easy to see from the definition that $R_{uu}(\tau) = R_{uu}^*(-\tau)$. Assume further that the Fourier transform of the signal's autocorrelation matrix function exists (and may contain impulses). This Fourier transform is called the *spectral density* of u , denoted $S_{uu}(j\omega)$:

$$S_{uu}(j\omega) := \int_{-\infty}^{\infty} R_{uu}(\tau) e^{-j\omega\tau} d\tau.$$

Thus $R_{uu}(\tau)$ can be obtained from $S_{uu}(j\omega)$ by performing an inverse Fourier transform:

$$R_{uu}(\tau) := \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{uu}(j\omega) e^{j\omega\tau} d\omega.$$

We will call signal $u(t)$ a *power signal* if the autocorrelation matrix $R_{uu}(\tau)$ exists and is finite for all τ , and moreover, if the power spectral density function $S_{uu}(j\omega)$ exists (note that $S_{uu}(j\omega)$ need not be bounded and may include impulses).

The power of the signal is defined as

$$\|u\|_{\mathcal{P}} = \sqrt{\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \|u(t)\|^2 dt} = \sqrt{\text{Trace}[R_{uu}(0)]}$$

where $\|\cdot\|$ is the usual Euclidean norm and the capital script $\mathcal{P}$ is used to differentiate this power semi-norm from the usual Lebesgue $\mathcal{L}_p$ norm. The set of all signals having finite power will be denoted by $\mathcal{P}$.

The power semi-norm of a signal can also be computed from its spectral density function

$$\|u\|_{\mathcal{P}}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{Trace}[S_{uu}(j\omega)] d\omega.$$

This expression implies that, if $u \in \mathcal{P}$, S_{uu} is strictly proper in the sense that $S_{uu}(\infty) = 0$. We note that if $u \in \mathcal{P}$ and $\|u(t)\|_{\infty} := \sup_t \|u(t)\| < \infty$, then $\|u\|_{\mathcal{P}} \leq \|u\|_{\infty}$. However, not every signal having finite ∞ -norm is a power signal since the limit in the definition may not exist. For example, let

$$u(t) = \begin{cases} 1 & 2^{2k} < t < 2^{2k+1}, \text{ for } k = 0, 1, 2, \dots \\ 0 & \text{otherwise.} \end{cases}$$

Then $\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T u^2 dt$ does not exist. Note also that power signals are persistent signals in time such as sines or cosines; clearly, a time-limited signal has zero power, as does an $\mathcal{L}_2$ signal. Thus $\|\cdot\|_{\mathcal{P}}$ is only a semi-norm, not a norm.

Now let G be a linear system transfer matrix with convolution kernel $g(t)$, input $u(t)$, and output $z(t)$. Then $R_{zz}(\tau) = g(\tau) * R_{uu}(\tau) * g^*(-\tau)$ and $S_{zz}(j\omega) = G(j\omega)S_{uu}(j\omega)G^*(j\omega)$. These properties are useful in establishing some input and output relationships in the next section.

A signal $u(t)$ is said to have bounded power spectral density if $\|S_{uu}(j\omega)\|_{\infty} < \infty$. The set of signals having bounded spectral density is denoted as

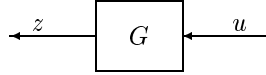
$$\mathcal{S} := \{u(t) \in \mathbb{R}^m : \|S_{uu}(j\omega)\|_{\infty} < \infty\}.$$

The quantity $\|u\|_s := \sqrt{\|S_{uu}(j\omega)\|_{\infty}}$ is called the spectral density norm of $u(t)$. The set $\mathcal{S}$ can be used to model signals with fixed spectral characteristics by passing white noise signals through a weighting filter. Similarly, $\mathcal{P}$ could be used to model signals whose spectrum is not known but which are finite in power.

4.5 Induced System Gains

Many control engineering problems involve keeping some system signals “small” under various conditions, for example, under a set of possible disturbances and system parameter variations. In this section we are interested in answering the following question: if we know how “big” the input (disturbance) is, how “big” is the output going to be for a given stable dynamical system?

Consider a q -input and p -output linear finite dimensional system as shown in the following diagram with input u , output z , and transfer matrix $G \in \mathcal{RH}_\infty$:



We will further assume that $G(s)$ is strictly proper, i.e., $G(\infty) = 0$ although most of the results derived here hold for the non-strictly proper case. In the time-domain an input-output model for such a system has the form of a convolution equation,

$$z = g * u$$

i.e.,

$$z(t) = \int_0^t g(t - \tau)u(\tau) d\tau$$

where the $p \times q$ real matrix $g(t)$ is the convolution kernel. Let the convolution kernel and the corresponding transfer matrix be partitioned as

$$g(t) = \begin{bmatrix} g_{11}(t) & \cdots & g_{1q}(t) \\ \vdots & & \vdots \\ g_{p1}(t) & \cdots & g_{pq}(t) \end{bmatrix} = \begin{bmatrix} g_1(t) \\ \vdots \\ g_p(t) \end{bmatrix},$$

$$G(s) = \begin{bmatrix} G_{11}(s) & \cdots & G_{1q}(s) \\ \vdots & & \vdots \\ G_{p1}(s) & \cdots & G_{pq}(s) \end{bmatrix} = \begin{bmatrix} G_1(s) \\ \vdots \\ G_p(s) \end{bmatrix}$$

where $g_i(t)$ is a q -dimensional row vector of the convolution kernel and $G_i(s)$ is a row vector of the transfer matrix. Now if G is considered as an operator from the input space to the output space, then a norm is induced on G , which, loosely speaking, measures the size of the output for a given input u . These norms can determine the achievable performance of the system for different classes of input signals

The various input/output relationships, given different classes of input signals, are summarized in two tables below. Table 4.1 summarizes the results for fixed input signals. Note that the $\delta(t)$ in this table denotes the unit impulse and $u_0 \in \mathbb{R}^q$ is a constant vector indicating the direction of the input signal.

We now prove Table 4.1.

Input $u(t)$	Output $z(t)$
$u(t) = u_0 \delta(t), u_0 \in \mathbb{R}^q$	$\ z\ _2 = \ Gu_0\ _2 = \ gu_0\ _2$
	$\ z\ _\infty = \ gu_0\ _\infty$
	$\ z\ _{\mathcal{P}} = 0$
$u(t) = u_0 \sin \omega_0 t, u_0 \in \mathbb{R}^q$	$\ z\ _2 = \infty$
	$\limsup_{t \rightarrow \infty} \max_i z_i(t) = \ G(j\omega_0)u_0\ _\infty = \max_i G_i(j\omega_0)u_0 $ $\limsup_{t \rightarrow \infty} \ z(t)\ = \ G(j\omega_0)u_0\ $
	$\ z\ _{\mathcal{P}} = \frac{1}{\sqrt{2}} \ G(j\omega_0)u_0\ $

Table 4.1: Output norms with fixed inputs

Proof.

$u = u_0 \delta(t)$: If $u(t) = u_0 \delta(t)$, then $z(t) = g(t)u_0$, so $\|z\|_2 = \|gu_0\|_2$ and $\|z\|_\infty = \|gu_0\|_\infty$. But by Parseval's relation, $\|gu_0\|_2 = \|Gu_0\|_2$. On the other hand,

$$\begin{aligned}
\|z\|_{\mathcal{P}}^2 &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T u_0^* g^*(t) g(t) u_0 \, dt \\
&\leq \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-\infty}^{\infty} \text{Trace}\{g^*(t)g(t)\} \, dt \|u_0\|^2 \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \|G\|_2^2 \|u_0\|^2 = 0.
\end{aligned}$$

$u = u_0 \sin \omega_0 t$: With the input $u(t) = u_0 \sin(\omega_0 t)$, the i -th output as $t \rightarrow \infty$ is

$$z_i(t) = |G_i(j\omega_0)u_0| \sin[\omega_0 t + \arg\{G_i(j\omega_0)u_0\}]. \quad (4.1)$$

The 2-norm of this signal is infinite as long as $G_i(j\omega_0)u_0 \neq 0$, i.e., the system's transfer function does not have a zero in every channel in the input direction at the frequency of excitation.

The amplitude of the sinusoid (4.1) equals $|G_i(j\omega_0)u_0|$. Hence

$$\limsup_{t \rightarrow \infty} \max_i |z_i(t)| = \max_i |G_i(j\omega_0)u_0| = \|G(j\omega_0)u_0\|_\infty$$

and

$$\limsup_{t \rightarrow \infty} \|z(t)\| = \sqrt{\sum_{i=1}^p |G_i(j\omega_0)u_0|^2} = \|G(j\omega_0)u_0\|.$$

Finally, let $\phi_i := \arg G_i(j\omega)u_0$. Then

$$\begin{aligned} \|z\|_{\mathcal{P}}^2 &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \sum_{i=1}^p |G_i(j\omega_0)u_0|^2 \sin^2(\omega_0 t + \phi_i) dt \\ &= \sum_{i=1}^p |G_i(j\omega_0)u_0|^2 \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \sin^2(\omega_0 t + \phi_i) dt \\ &= \sum_{i=1}^p |G_i(j\omega_0)u_0|^2 \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \frac{1 - \cos 2(\omega_0 t + \phi_i)}{2} dt \\ &= \frac{1}{2} \sum_{i=1}^p |G_i(j\omega_0)u_0|^2 = \frac{1}{2} \|G(j\omega_0)u_0\|^2. \end{aligned}$$

□

It is interesting to see what this table signifies from the control point of view. To focus our discussion, let us assume that $u(t) = u_0 \sin \omega_0 t$ is a disturbance or a command signal on a feedback system, and $z(t)$ is the tracking error. Then we say that the system has good tracking behavior if $z(t)$ is small in some sense, for instance, $\limsup_{t \rightarrow \infty} \|z(t)\|$ is small. Note that

$$\limsup_{t \rightarrow \infty} \|z(t)\| = \|G(j\omega_0)u_0\|$$

for any given ω_0 and $u_0 \in \mathbb{R}^q$. Now if we want to track signals from various channels, that is if u_0 can be chosen to be any direction, then we would require that $\bar{\sigma}(G(j\omega_0))$ be small. Furthermore, if, in addition, we want to track signals of many different frequencies, we then would require that $\bar{\sigma}(G(j\omega_0))$ be small at all those frequencies. This interpretation enables us to consider the control system in the frequency domain even though the specifications are given in the time domain.

Table 4.2 lists the maximum possible system gains when the input signal u is not required to be a fixed signal; instead it can be any signal in a unit ball in some function space.

Now we give a proof for Table 4.2. Note that the first row ($\mathcal{L}_2 \mapsto \mathcal{L}_2$) has been shown in Section 4.3.

Input $u(t)$	Output $z(t)$	Signal Norms	Induced Norms
$\mathcal{L}_2$	$\mathcal{L}_2$	$\ u\ _2^2 = \int_0^\infty \ u\ ^2 dt$	$\ G\ _\infty$
$\mathcal{S}$	$\mathcal{S}$	$\ u\ _{\mathcal{S}}^2 = \ S_{uu}\ _\infty$	$\ G\ _\infty$
$\mathcal{S}$	$\mathcal{P}$	$\ u\ _{\mathcal{P}}^2 = \frac{1}{2\pi} \int_{-\infty}^\infty \text{Trace}\{S_{uu}(j\omega)\} d\omega$	$\ G\ _2$
$\mathcal{P}$	$\mathcal{P}$	$\ u\ _{\mathcal{P}}^2 = \frac{1}{2\pi} \int_{-\infty}^\infty \text{Trace}\{S_{uu}(j\omega)\} d\omega$	$\ G\ _\infty$
$\mathcal{L}_\infty$	$\mathcal{L}_\infty$	$\ u\ _\infty = \sup_t \max_i u_i(t) $	$\max_i \ g_i\ _1$
		$\ u\ _\infty = \sup_t \ u(t)\ $	$\leq \int_0^\infty \ g(t)\ dt$

Table 4.2: Induced System Gains

Proof.

$\mathcal{S} \mapsto \mathcal{S}$: If $u \in \mathcal{S}$, then

$$S_{zz}(j\omega) = G(j\omega)S_{uu}(j\omega)G^*(j\omega).$$

Now suppose

$$\overline{\sigma}[G(j\omega_0)] = \|G\|_\infty$$

and take a signal u such that $S_{uu}(j\omega_0) = I$. Then

$$\|S_{zz}(j\omega)\|_\infty = \|G\|_\infty^2.$$

$\mathcal{S} \mapsto \mathcal{P}$: By definition, we have

$$\|z\|_{\mathcal{P}}^2 = \frac{1}{2\pi} \int_{-\infty}^\infty \text{Trace}\{G(j\omega)S_{uu}(j\omega)G^*(j\omega)\} d\omega.$$

Now let u be white with unit spectral density, i.e., $S_{uu} = I$. Then

$$\|z\|_{\mathcal{P}}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{Trace}\{G(j\omega)G^*(j\omega)\} d\omega = \|G\|_2^2.$$

$\mathcal{P} \mapsto \mathcal{P}$:

If u is a power signal, then

$$\|z\|_{\mathcal{P}}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{Trace}\{G(j\omega)S_{uu}(j\omega)G^*(j\omega)\} d\omega,$$

and immediately, we get that

$$\|z\|_{\mathcal{P}} \leq \|G\|_{\infty} \|u\|_{\mathcal{P}}.$$

To achieve the equality, assume that ω_0 is such that

$$\overline{\sigma}[G(j\omega_0)] = \|G\|_{\infty}$$

and denote the singular value decomposition of $G(j\omega_0)$ by

$$G(j\omega_0) = \overline{\sigma}u_1(j\omega_0)v_1^*(j\omega_0) + \sum_{i=2}^r \sigma_i u_i(j\omega_0)v_i^*(j\omega_0)$$

where r is the rank of $G(j\omega_0)$ and u_i, v_i have unit length.

If $\omega_0 < \infty$, write $v_1(j\omega_0)$ as

$$v_1(j\omega) = \begin{bmatrix} \alpha_1 e^{j\theta_1} \\ \alpha_2 e^{j\theta_2} \\ \vdots \\ \alpha_q e^{j\theta_q} \end{bmatrix}$$

where $\alpha_i \in \mathbb{R}$ is such that $\theta_i \in [-\pi, 0)$ and q is the column dimension of G . Now let $\beta_i \geq 0$ be such that

$$\theta_i = \angle \left(\frac{\beta_i - j\omega_0}{\beta_i + j\omega_0} \right)$$

and let the input u be generated from passing $\hat{u}$ through a filter:

$$u(t) = \begin{bmatrix} \alpha_1 \frac{\beta_1 - s}{\beta_1 + s} \\ \alpha_2 \frac{\beta_2 - s}{\beta_2 + s} \\ \vdots \\ \alpha_q \frac{\beta_q - s}{\beta_q + s} \end{bmatrix} \hat{u}(t)$$

where

$$\hat{u}(t) = \sqrt{2} \sin(\omega_o t).$$

Then $R_{\hat{u}\hat{u}}(\tau) = \cos(\omega_o \tau)$, so

$$\|\hat{u}\|_{\mathcal{P}} = R_{\hat{u}\hat{u}}(0) = 1.$$

Also,

$$S_{\hat{u}\hat{u}}(j\omega) = \pi [\delta(\omega - \omega_o) + \delta(\omega + \omega_o)].$$

Then

$$S_{uu}(j\omega) = \begin{bmatrix} \alpha_1 \frac{\beta_1 - j\omega}{\beta_1 + j\omega} \\ \alpha_2 \frac{\beta_2 - j\omega}{\beta_2 + j\omega} \\ \vdots \\ \alpha_q \frac{\beta_q - j\omega}{\beta_q + j\omega} \end{bmatrix} S_{\hat{u}\hat{u}}(j\omega) \begin{bmatrix} \alpha_1 \frac{\beta_1 - j\omega}{\beta_1 + j\omega} \\ \alpha_2 \frac{\beta_2 - j\omega}{\beta_2 + j\omega} \\ \vdots \\ \alpha_q \frac{\beta_q - j\omega}{\beta_q + j\omega} \end{bmatrix}^*$$

and it is easy to show

$$\|u\|_{\mathcal{P}} = 1.$$

Finally,

$$\begin{aligned} \|z\|_{\mathcal{P}}^2 &= \frac{1}{2} \overline{\sigma} [G(j\omega_o)]^2 + \frac{1}{2} \overline{\sigma} [G(-j\omega_o)]^2 \\ &= \overline{\sigma} [G(-j\omega_o)]^2 \\ &= \|G\|_{\infty}^2. \end{aligned}$$

Similarly, if $\omega_o = \infty$, then $G(j\omega)$ is real. The conclusion follows by letting $u = v_1 \hat{u}(t)$ and $\omega_o \rightarrow \infty$.

$\mathcal{L}_{\infty} \mapsto \mathcal{L}_{\infty}$:

1. First of all, $\max_i \|g_i\|_1$ is an upper bound on the ∞ -norm/ ∞ -norm system gain:

$$\begin{aligned} |z_i(t)| &= \left| \int_0^t g_i(\tau) u(t - \tau) d\tau \right| \leq \int_0^t |g_i(\tau) u(t - \tau)| d\tau \\ &\leq \int_0^t \sum_{j=1}^q |g_{ij}(\tau)| d\tau \|u\|_{\infty} \leq \int_0^{\infty} \sum_{j=1}^q |g_{ij}(\tau)| d\tau \|u\|_{\infty} \\ &= \|g_i\|_1 \|u\|_{\infty} \end{aligned}$$

and

$$\begin{aligned} \|z\|_{\infty} &:= \max_i \sup_t |z_i(t)| \\ &\leq \max_i \|g_i\|_1 \|u\|_{\infty}. \end{aligned}$$

That $\max_i \|g_i\|_1$ is the least upper bound can be seen as follows: Assume that the maximum $\max_i \|g_i\|_1$ is achieved for $i = 1$. Let t_0 be given and set

$$u(t_0 - \tau) := \text{sign}(g_1^*(\tau)), \quad \forall \tau.$$

Note that since $g_1(\tau)$ is a vector function, the sign function $\text{sign}(g_1^*(\tau))$ is a component-wise operation. Then $\|u\|_\infty = 1$ and

$$\begin{aligned} z_1(t_0) &= \int_0^{t_0} g_1(\tau) u(t_0 - \tau) d\tau \\ &= \int_0^{t_0} \sum_{j=1}^q |g_{1j}(\tau)| d\tau \\ &= \|g_1\|_1 - \int_{t_0}^\infty \sum_{j=1}^q |g_{1j}(\tau)| d\tau. \end{aligned}$$

Hence, let $t_0 \rightarrow \infty$, and we have $\|z\|_\infty = \|g_1\|_1$.

2. If $\|u(t)\|_\infty := \sup_t \|u(t)\|$, then

$$\begin{aligned} \|z(t)\| &= \left\| \int_0^t g(\tau) u(t - \tau) d\tau \right\| \\ &\leq \int_0^t \|g(\tau)\| \|u(t - \tau)\| d\tau \\ &\leq \int_0^t \|g(\tau)\| d\tau \|u\|_\infty. \end{aligned}$$

And, therefore, $\|z\|_\infty \leq \int_0^\infty \|g(\tau)\| d\tau \|u\|_\infty$.

□

Next we shall derive some simple and useful bounds for the $\mathcal{H}_\infty$ norm and the $\mathcal{L}_1$ norm of a stable system. Suppose

$$G(s) = \left[\frac{A}{C} \middle| \frac{B}{0} \right] \in \mathcal{RH}_\infty$$

is a balanced realization, i.e., there exists

$$\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_n) \geq 0$$

with $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n \geq 0$, such that

$$A\Sigma + \Sigma A^* + BB^* = 0 \quad A^*\Sigma + \Sigma A + C^*C = 0.$$

Then we have the following theorem.

Theorem 4.5

$$\sigma_1 \leq \|G\|_\infty \leq \int_0^\infty \|g(t)\| dt \leq 2 \sum_{i=1}^n \sigma_i$$

where $g(t) = Ce^{At}B$.

Remark 4.3 It should be clear that the inequalities stated in the theorem do not depend on a particular state space realization of $G(s)$. However, use of the balanced realization does make the proof simple. $\heartsuit$

Proof. The inequality $\sigma_1 \leq \|G\|_\infty$ follows from the Nehari Theorem of Chapter 8. We will now show the other inequalities. Since

$$G(s) := \int_0^\infty g(t)e^{-st}dt, \quad \operatorname{Re}(s) > 0,$$

by the definition of $\mathcal{H}_\infty$ norm, we have

$$\begin{aligned} \|G\|_\infty &= \sup_{\operatorname{Re}(s) > 0} \left\| \int_0^\infty g(t)e^{-st}dt \right\| \\ &\leq \sup_{\operatorname{Re}(s) > 0} \int_0^\infty \|g(t)e^{-st}\| dt \\ &\leq \int_0^\infty \|g(t)\| dt. \end{aligned}$$

To prove the last inequality, let u_i be the i^{th} unit vector. Then

$$u_i^* u_j = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \text{and} \quad \sum_{i=1}^n u_i u_i^* = I.$$

Define $\gamma_i(t) = u_i^* e^{At/2} B$ and $\psi_i(t) = C e^{At/2} u_i$. It is easy to verify that

$$\|\gamma_i\|_2^2 = \int_0^\infty u_i^* e^{At/2} B B^* e^{A^* t/2} u_i dt = 2u_i^* \Sigma u_i = 2\sigma_i$$

$$\|\psi_i\|_2^2 = \int_0^\infty u_i^* e^{A^* t/2} C^* C e^{At/2} u_i dt = 2u_i^* \Sigma u_i = 2\sigma_i.$$

Using these two equalities, we have

$$\begin{aligned} \int_0^\infty \|g(t)\| dt &= \int_0^\infty \left\| \sum_{i=1}^n \psi_i \gamma_i \right\| dt \leq \sum_{i=1}^n \int_0^\infty \|\psi_i \gamma_i\| dt \\ &\leq \sum_{i=1}^n \|\psi_i\|_2 \|\gamma_i\|_2 \leq 2 \sum_{i=1}^n \sigma_i. \end{aligned}$$

□

It should be clear from the above two tables that many system performance criteria can be stipulated as requiring a certain closed loop transfer matrix have small $\mathcal{H}_2$ norm or $\mathcal{H}_\infty$ norm or L_1 norm. Moreover, if L_1 performance is satisfied, then the $\mathcal{H}_\infty$ norm performance is also satisfied. We will be most interested in $\mathcal{H}_2$ and $\mathcal{H}_\infty$ performance in this book.

4.6 Computing $\mathcal{L}_2$ and $\mathcal{H}_2$ Norms

Let $G(s) \in \mathcal{L}_2$ and recall that the $\mathcal{L}_2$ norm of G is defined as

$$\begin{aligned} \|G\|_2 &:= \sqrt{\frac{1}{2\pi} \int_{-\infty}^{\infty} \text{Trace}\{G^*(j\omega)G(j\omega)\} d\omega} \\ &= \|g\|_2 \\ &= \sqrt{\int_{-\infty}^{\infty} \text{Trace}\{g^*(t)g(t)\} dt} \end{aligned}$$

where $g(t)$ denotes the convolution kernel of G .

It is easy to see that the $\mathcal{L}_2$ norm defined above is finite iff the transfer matrix G is strictly proper, i.e., $G(\infty) = 0$. Hence, we will generally assume that the transfer matrix is strictly proper whenever we refer to the $\mathcal{L}_2$ norm of G (of course, this also applies to $\mathcal{H}_2$ functions). One straightforward way of computing the $\mathcal{L}_2$ norm is to use contour integral. Suppose G is strictly proper, and then we have

$$\begin{aligned} \|G\|_2^2 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{Trace}\{G^*(j\omega)G(j\omega)\} d\omega \\ &= \frac{1}{2\pi j} \oint \text{Trace}\{G^\sim(s)G(s)\} ds. \end{aligned}$$

The last integral is a contour integral along the imaginary axis, and around an infinite semi-circle in the left half-plane; the contribution to the integral from this semi-circle equals zero because G is strictly proper. By the residue theorem, $\|G\|_2^2$ equals the sum of the residues of $\text{Trace}\{G^\sim(s)G(s)\}$ at its poles in the left half-plane.

Although $\|G\|_2$ can, in principle, be computed from its definition or from the method suggested above, it is useful in many applications to have alternative characterizations and to take advantage of the state space representations of G . The computation of a $\mathcal{RH}_2$ transfer matrix norm is particularly simple.

Lemma 4.6 *Consider a transfer matrix*

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array} \right]$$

with A stable. Then we have

$$\|G\|_2^2 = \text{trace}(B^* L_o B) = \text{trace}(C L_c C^*) \quad (4.2)$$

where L_o and L_c are observability and controllability Gramians which can be obtained from the following Lyapunov equations

$$A L_c + L_c A^* + B B^* = 0 \quad A^* L_o + L_o A + C^* C = 0.$$

Proof. Since G is stable, we have

$$g(t) = \mathcal{L}^{-1}(G) = \begin{cases} C e^{A t} B, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

and

$$\begin{aligned} \|G\|_2^2 &= \int_0^\infty \text{Trace}\{g^*(t)g(t)\} dt = \int_0^\infty \text{Trace}\{g(t)g(t)^*\} dt \\ &= \int_0^\infty \text{Trace}\{B^* e^{A^* t} C^* C e^{A t} B\} dt = \int_0^\infty \text{Trace}\{C e^{A t} B B^* e^{A^* t} C^*\} dt. \end{aligned}$$

The lemma follows from the fact that the controllability Gramian of (A, B) and the observability Gramian of (C, A) can be represented as

$$L_o = \int_0^\infty e^{A^* t} C^* C e^{A t} dt, \quad L_c = \int_0^\infty e^{A t} B B^* e^{A^* t} dt,$$

which can also be obtained from

$$A L_c + L_c A^* + B B^* = 0 \quad A^* L_o + L_o A + C^* C = 0.$$

□

To compute the $\mathcal{L}_2$ norm of a rational transfer function, $G(s) \in \mathcal{L}_2$, using state space approach. Let $G(s) = [G(s)]_+ + [G(s)]_-$ with $G_+ \in \mathcal{RH}_2$ and $G_- \in \mathcal{RH}_2^\perp$. Then

$$\|G\|_2^2 = \|[G(s)]_+\|_2^2 + \|[G(s)]_-\|_2^2$$

where $\|[G(s)]_+\|_2$ and $\|[G(s)]_-\|_2 = \|[G(-s)]_+\|_2$ can be computed using the above lemma.

Still another useful characterization of the $\mathcal{H}_2$ norm of G is in terms of hypothetical input-output experiments. Let e_i denote the i^{th} standard basis vector of $\mathbb{R}^m$ where m is the input dimension of the system. Apply the impulsive input $\delta(t)e_i$ ($\delta(t)$ is the unit impulse) and denote the output by $z_i(t)(= g(t)e_i)$. Assume $D = 0$, and then $z_i \in \mathcal{L}_{2+}$ and

$$\|G\|_2^2 = \sum_{i=1}^m \|z_i\|_2^2.$$

Note that this characterization of the $\mathcal{H}_2$ norm can be appropriately generalized for nonlinear time varying systems, see Chen and Francis [1992] for an application of this norm in sampled-data control.

4.7 Computing $\mathcal{L}_\infty$ and $\mathcal{H}_\infty$ Norms

We shall first consider, as in the $\mathcal{L}_2$ case, how to compute the ∞ norm of an $\mathcal{L}_\infty$ transfer matrix. Let $G(s) \in \mathcal{L}_\infty$ and recall that the $\mathcal{L}_\infty$ norm of a transfer function G is defined as

$$\|G\|_\infty := \operatorname{ess\,sup}_\omega \bar{\sigma}\{G(j\omega)\}.$$

The computation of the $\mathcal{L}_\infty$ norm of G is complicated and requires a search. A control engineering interpretation of the infinity norm of a scalar transfer function G is the distance in the complex plane from the origin to the farthest point on the Nyquist plot of G , and it also appears as the peak value on the Bode magnitude plot of $|G(j\omega)|$. Hence the ∞ of a transfer function can in principle be obtained graphically.

To get an estimate, set up a fine grid of frequency points,

$$\{\omega_1, \dots, \omega_N\}.$$

Then an estimate for $\|G\|_\infty$ is

$$\max_{1 \leq k \leq N} \bar{\sigma}\{G(j\omega_k)\}.$$

This value is usually read directly from a Bode singular value plot. The $\mathcal{L}_\infty$ norm can also be computed in state space if G is rational.

Lemma 4.7 *Let $\gamma > 0$ and*

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RL}_\infty. \quad (4.3)$$

Then $\|G\|_\infty < \gamma$ if and only if $\bar{\sigma}(D) < \gamma$ and H has no eigenvalues on the imaginary axis where

$$H := \left[\begin{array}{cc} A + BR^{-1}D^*C & BR^{-1}B^* \\ -C^*(I + DR^{-1}D^*)C & -(A + BR^{-1}D^*C)^* \end{array} \right] \quad (4.4)$$

*and $R = \gamma^2 I - D^*D$.*

Proof. Let $\Phi(s) = \gamma^2 I - G^*(s)G(s)$. Then it is clear that $\|G\|_\infty < \gamma$ if and only if $\Phi(j\omega) > 0$ for all $\omega \in \mathbb{R}$. Since $\Phi(\infty) = R > 0$ and since $\Phi(j\omega)$ is a continuous function of ω , $\Phi(j\omega) > 0$ for all $\omega \in \mathbb{R}$ if and only if $\Phi(j\omega)$ is nonsingular for all $\omega \in \mathbb{R} \cup \{\infty\}$, i.e., $\Phi(s)$ has no imaginary axis zero. Equivalently, $\Phi^{-1}(s)$ has no imaginary axis pole. It is easy to compute by some simple algebra that

$$\Phi^{-1}(s) = \left[\begin{array}{c|c} H & \left[\begin{array}{c} BR^{-1} \\ -C^*DR^{-1} \end{array} \right] \\ \hline \left[\begin{array}{cc} R^{-1}D^*C & R^{-1}B^* \end{array} \right] & R^{-1} \end{array} \right].$$

Thus the conclusion follows if the above realization has neither uncontrollable modes nor unobservable modes on the imaginary axis. Assume that $j\omega_0$ is an eigenvalue of H but not a pole of $\Phi^{-1}(s)$. Then $j\omega_0$ must be either an unobservable mode of $([R^{-1}D^*C \ R^{-1}B^*], H)$ or an uncontrollable mode of $(H, [BR^{-1} \ -C^*DR^{-1}])$. Now suppose $j\omega_0$ is an unobservable mode of $([R^{-1}D^*C \ R^{-1}B^*], H)$. Then there exists an $x_0 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \neq 0$ such that

$$Hx_0 = j\omega_0 x_0, \quad [R^{-1}D^*C \ R^{-1}B^*] x_0 = 0.$$

These equations can be simplified to

$$\begin{aligned} (j\omega_0 I - A)x_1 &= 0 \\ (j\omega_0 I + A^*)x_2 &= -C^*C x_1 \\ D^*C x_1 + B^*x_2 &= 0. \end{aligned}$$

Since A has no imaginary axis eigenvalues, we have $x_1 = 0$ and $x_2 = 0$. This contradicts our assumption, and hence the realization has no unobservable modes on the imaginary axis.

Similarly, a contradiction will also be arrived if $j\omega_0$ is assumed to be an uncontrollable mode of $(H, [BR^{-1} \ -C^*DR^{-1}])$. $\square$

Bisection Algorithm

Lemma 4.7 suggests the following bisection algorithm to compute $\mathcal{RL}_\infty$ norm:

- (a) select an upper bound γ_u and a lower bound γ_l such that $\gamma_l \leq \|G\|_\infty \leq \gamma_u$;
- (b) if $(\gamma_u - \gamma_l)/\gamma_l \leq \text{specified level}$, stop; $\|G\| \approx (\gamma_u + \gamma_l)/2$. Otherwise go to next step;
- (c) set $\gamma = (\gamma_l + \gamma_u)/2$;
- (d) test if $\|G\|_\infty < \gamma$ by calculating the eigenvalues of H for the given γ ;
- (e) if H has an eigenvalue on $j\mathbb{R}$ set $\gamma_l = \gamma$; otherwise set $\gamma_u = \gamma$; go back to step (b).

Of course, the above algorithm applies to $\mathcal{H}_\infty$ norm computation as well. Thus $\mathcal{L}_\infty$ norm computation requires a search, over either γ or ω , in contrast to $\mathcal{L}_2$ ($\mathcal{H}_2$) norm computation, which does not. A somewhat analogous situation occurs for constant matrices with the norms $\|M\|_2^2 = \text{trace}(M^*M)$ and $\|M\|_\infty = \bar{\sigma}[M]$. In principle, $\|M\|_2^2$ can be computed exactly with a finite number of operations, as can the test for whether $\bar{\sigma}(M) < \gamma$ (e.g. $\gamma^2 I - M^*M > 0$), but the value of $\bar{\sigma}(M)$ cannot. To compute $\bar{\sigma}(M)$, we must use some type of iterative algorithm.

Remark 4.4 It is clear that $\|G\|_\infty < \gamma$ iff $\|\gamma^{-1}G\|_\infty < 1$. Hence, there is no loss of generality to assume $\gamma = 1$. This assumption will often be made in the remainder of this book. It is also noted that there are other fast algorithms to carry out the above norm computation; nevertheless, this bisection algorithm is the simplest. $\heartsuit$

The $\mathcal{H}_\infty$ norm of a stable transfer function can also be estimated experimentally using the fact that the $\mathcal{H}_\infty$ norm of a stable transfer function is the maximum magnitude of the steady-state response to all possible unit amplitude sinusoidal input signals.

4.8 Notes and References

The basic concept of function spaces presented in this chapter can be found in any standard functional analysis textbook, for instance, Naylor and Sell [1982] and Gohberg and Goldberg [1981]. The system theoretical interpretations of the norms and function spaces can be found in Desoer and Vidyasagar [1975]. The bisection $\mathcal{H}_\infty$ norm computational algorithm is first developed in Boyd, Balakrishnan, and Kabamba [1989]. A more efficient $\mathcal{L}_\infty$ norm computational algorithm is presented in Bruinsma and Steinbuch [1990].